

DS_A2_Q2

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2.a

Consider the probability that S_j^* contains none of $X_1, \dots, X_\ell$ is

$$P(X_1, \dots, X_\ell \cap S_j^* = \emptyset)$$

and the probability that a single observation in S_j^* is none of $X_1, \dots, X_\ell$ is

$$\frac{n - \ell}{n}$$

Since we have n observations in S_j^* . Thus, the probability is

$$P(X_1, \dots, X_\ell \cap S_j^* = \emptyset) = \left(\frac{n - \ell}{n} \right)^n$$

2.b

Since we have k observations in S_j^* are from $X_1, \dots, X_\ell$, the sum of these observations is

$$\sum_{i=1}^k X_i = kE[X_i] = k(\theta - \phi)$$

And the left $n - k$ observations are from $X_{\ell+1}, \dots, X_n$, the sum of these observations is

$$\sum_{i=k+1}^n X_i = (n - k)E[X_i] = (n - k)\theta$$

Thus, the sum of all observations is

$$\sum_{i=1}^n X_i = k(\theta - \phi) + (n - k)\theta = n\theta - k\phi,$$

which determines that

$$\bar{X}_j = \theta - \frac{k\phi}{n}$$

2.c

As $\bar{X}_j$ is an unbiased estimator of θ , we have

$$E[\bar{X}_j] = \theta$$

According to 2.b, we have,

$$E[\bar{X}_j] = \theta - \frac{k\phi}{n} = \theta \Rightarrow \frac{k\phi}{n} = 0$$

which determines that

$$k = 0.$$

2.d

According to the definition of expected bias and 2.b, we have

$$\begin{aligned}\text{Bias}(\bar{X}_j) &= E[\bar{X}_j] - \theta \\ &= \theta - \frac{k\phi}{n} - \theta \\ &= -\frac{k\phi}{n}\end{aligned}$$